Chapter 18: Corrosion and Degradation of Materials

ISSUES TO ADDRESS...

- How does corrosion occur?
- Which metals are most likely to corrode?
- What environmental parameters affect corrosion rate?
- How do we prevent or control corrosion?

THE COST OF CORROSION

- Corrosion:
  -- the destructive electrochemical attack of a material.
  -- Ex: Rusting of automobiles and other equipment



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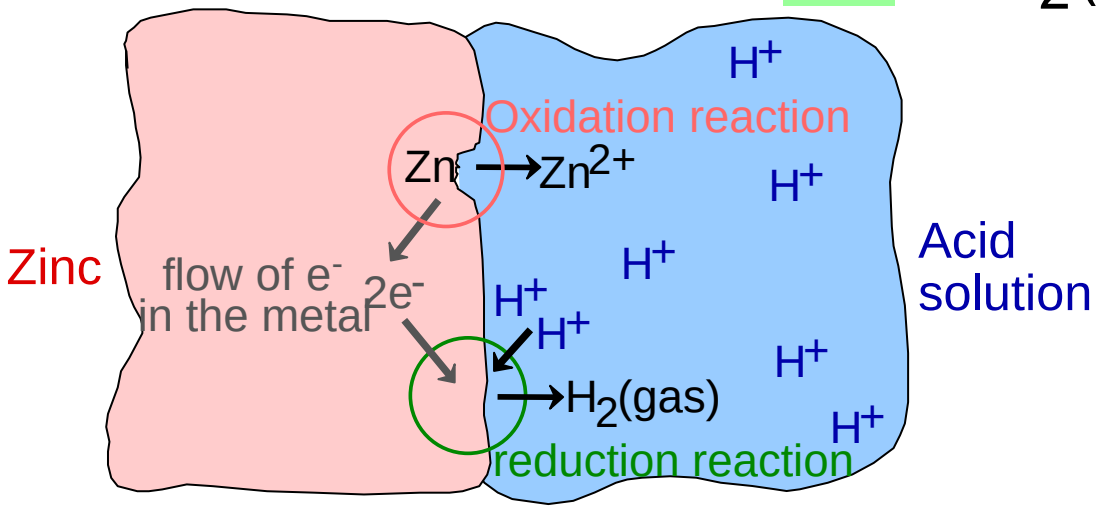
- Cost:
  -- 4 to 5% of the Gross National Product (GNP)*
  -- in the U.S. this amounts to just over $400 billion/yr**

* H.H. Uhlig and W.R. Revie, *Corrosion and Corrosion Control: An Introduction to Corrosion Science and Engineering*, 3rd ed., John Wiley and Sons, Inc., 1985.
**Economic Report of the President (1998).

ELECTROCHEMICAL CORROSION

Ex: consider the corrosion of zinc in an acid solution

- Two reactions are necessary:
  - -- oxidation reaction: $Zn \rightarrow Zn^{2+} + 2e^-$
  - -- reduction reaction: $2H^+ + 2e^- \rightarrow H_2(gas)$



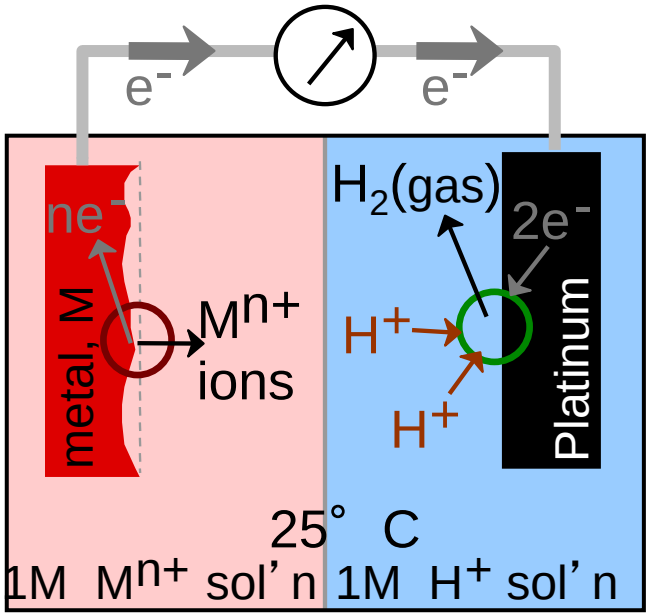
Adapted from Fig. 18.1, *Callister & Rethwisch 9e*. (From M. G. Fontana, *Corrosion Engineering*, 3rd edition. Copyright © 1986 by McGraw-Hill Book Company. Reproduced with permission.)

- Other reduction reactions in solutions with dissolved oxygen:
  - -- acidic solution: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
  - -- neutral or basic solution: $O_2 + 2H_2O + 4e^- \rightarrow 4(OH)^-$

STANDARD HYDROGEN ELECTRODE

- Two outcomes:

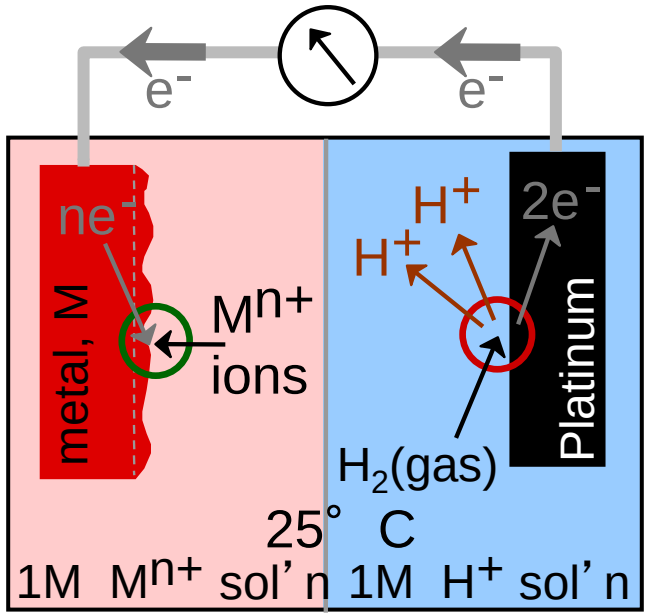
-- Corrosion



-- Metal is the anode (-)

$V^{o}_{metal} < 0$ (relative to Pt)

-- Electrodeposition



-- Metal is the cathode (+)

$V^{o}_{metal} > 0$ (relative to Pt)

Standard Electrode Potential

Adapted from Fig. 18.2, *Callister & Rethwisch 9e*.

STANDARD EMF SERIES

- EMF series

more cathodic

more anodic

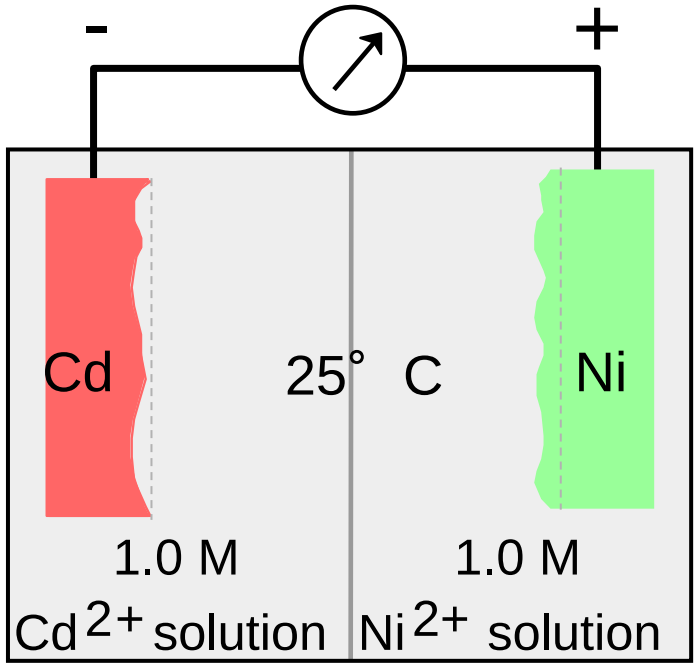
metal	V^o_{metal}
Au	+1.420 V
Cu	+0.340
Pb	- 0.126
Sn	- 0.136
Ni	- 0.250
Co	- 0.277
Cd	- 0.403
Fe	- 0.440
Cr	- 0.744
Zn	- 0.763
Al	- 1.662
Mg	- 2.363
Na	- 2.714
K	- 2.924

$\Delta V^o =$ 0.153V

Data based on Table 18.1, *Callister 9e*.

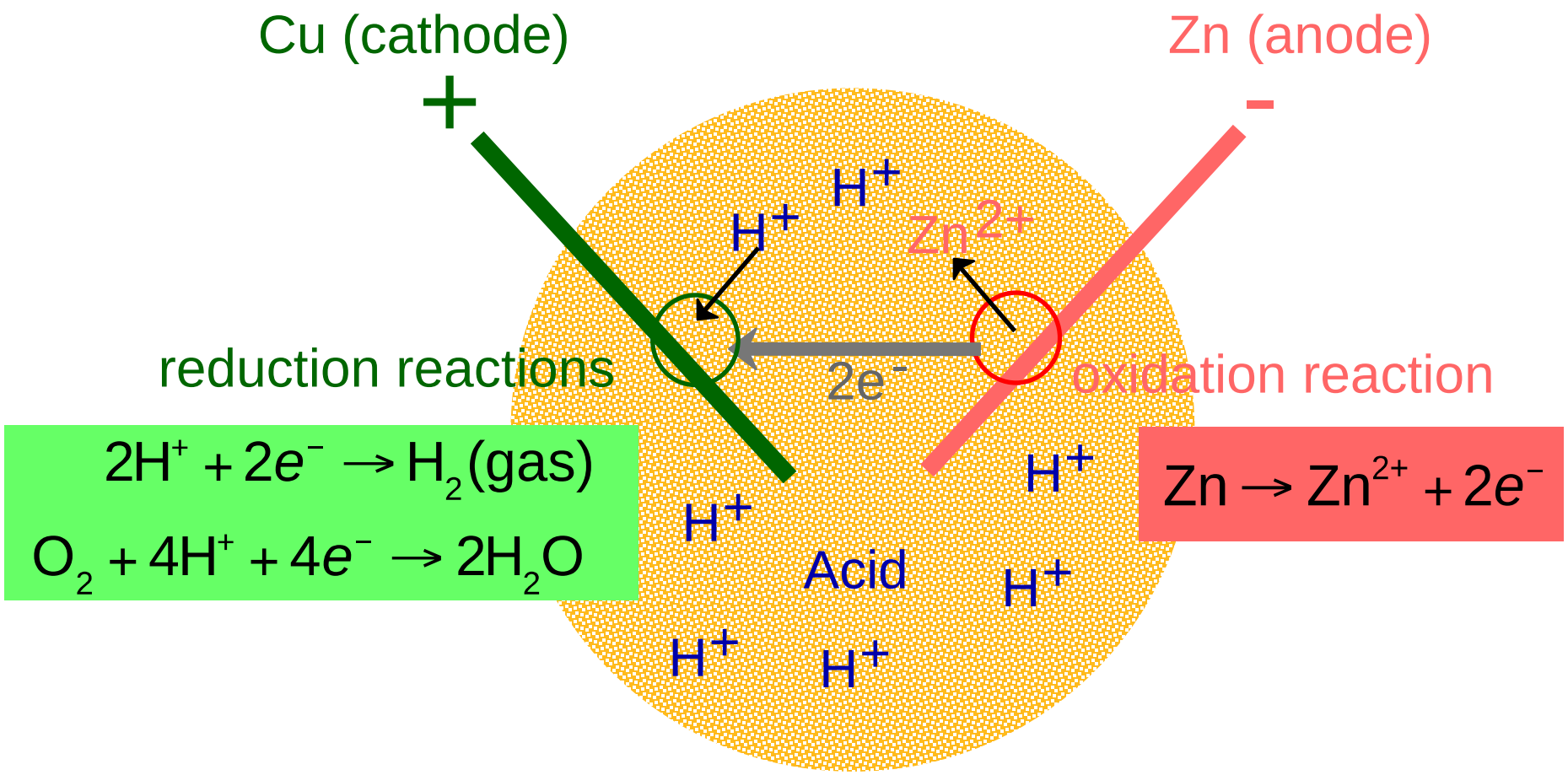
- Metal with smaller V^o_{metal} corrodes.
- Ex: Cd-Ni cell

$V^o_{Cd} < V^o_{Ni}$ $\therefore$ Cd corrodes



Adapted from Fig. 18.2, *Callister & Rethwisch 9e*.

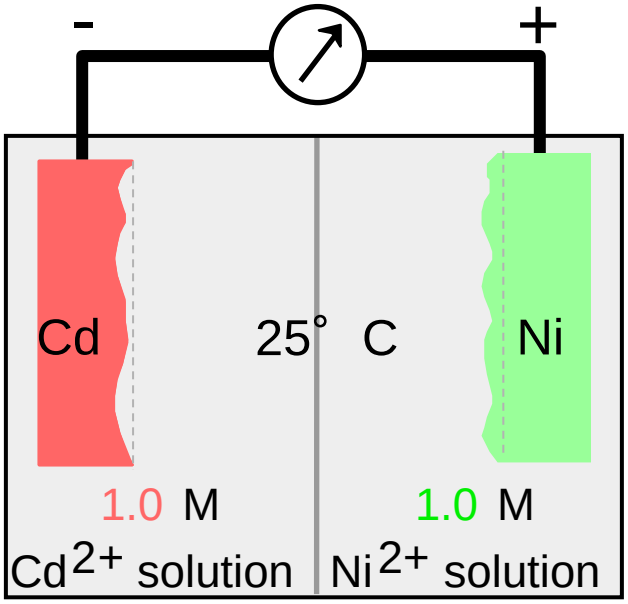
CORROSION IN A GRAPEFRUIT



EFFECT OF SOLUTION CONCENTRATION AND TEMPERATURE

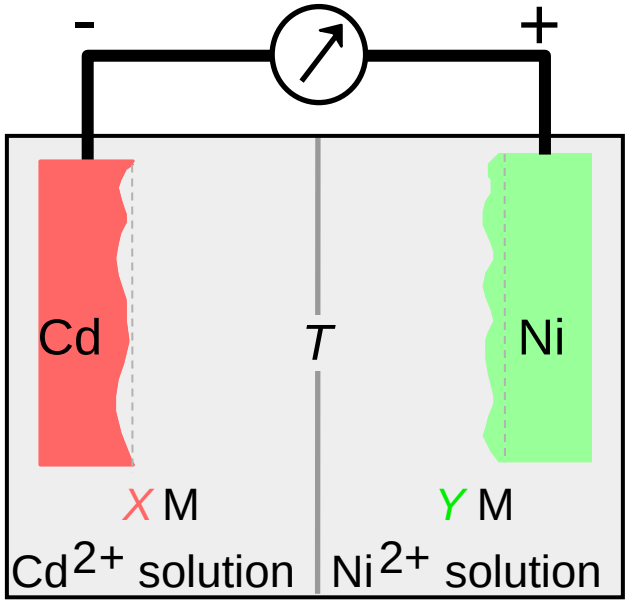
- Ex: Cd-Ni cell with standard 1 M solutions

$$V^{o}_{Ni} - V^{o}_{Cd} = 0.153\ V$$



- Ex: Cd-Ni cell with non-standard solutions

$$V_{Ni} - V_{Cd} = V^{o}_{Ni} - V^{o}_{Cd} - \frac{RT}{nF}\ln\frac{X}{Y}$$



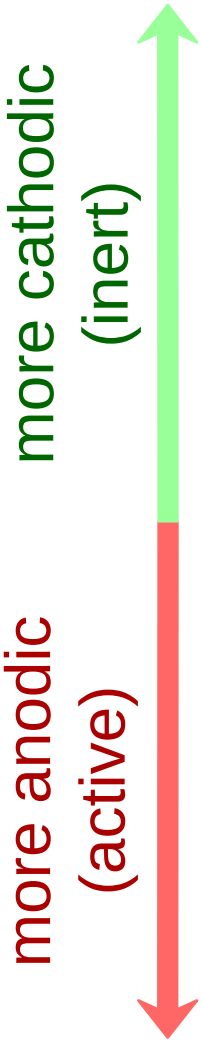
n = #e- per unit oxid/red reaction (= 2 here)

F = Faraday's constant = 96,500 C/mol.

- Reduce V_{Ni} - V_{Cd} by
  - -- increasing *X*
  - -- decreasing *Y*
  - -- increasing *T*

GALVANIC SERIES

- Ranking of the reactivity of metals/alloys in seawater



Platinum
Gold
Graphite
Titanium
Silver
316 Stainless Steel (passive)
Nickel (passive)
Copper
Nickel (active)
Tin
Lead
316 Stainless Steel (active)
Iron/Steel
Aluminum Alloys
Cadmium
Zinc
Magnesium

Table 18.2, *Callister & Rethwisch 9e*.
Source is M.G. Fontana, *Corrosion Engineering*, 3rd ed., McGraw-Hill Book Company, 1986. Reprinted with permission)

FORMS OF CORROSION

- **Uniform Attack**
  Oxidation & reduction reactions occur uniformly over surfaces.
- **Selective Leaching**
  Preferred corrosion of one element/constituent [e.g., Zn from brass (Cu-Zn)].
- **Intergranular**
  Corrosion along grain boundaries, often where precip. particles form.

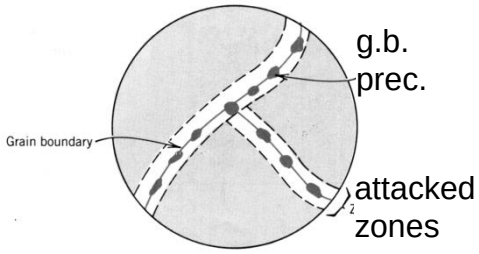


Fig. 18.18, *Callister & Rethwisch 9e*.

- **Stress corrosion**
  Corrosion at crack tips when a tensile stress is present.

Forms of corrosion

- **Galvanic**
  Dissimilar metals are physically joined in the presence of an electrolyte. The more anodic metal corrodes.

- **Erosion-corrosion**
  Combined chemical attack and mechanical wear (e.g., pipe elbows).
- **Pitting**
  Downward propagation of small pits and holes.

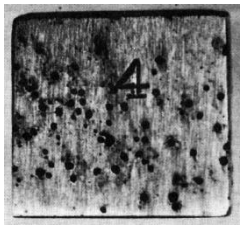


Fig. 18.17, *Callister & Rethwisch 9e*. (From M.G. Fontana, *Corrosion Engineering*, 3rd ed., McGraw-Hill Book Company, 1986.)

- **Crevice** Narrow and confined spaces.

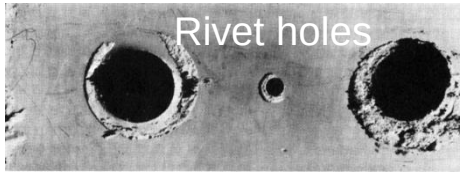
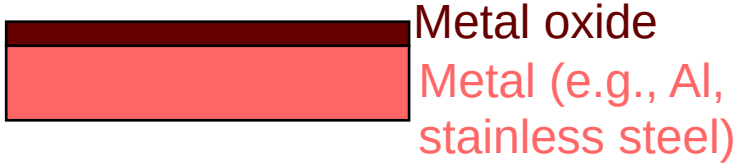


Fig. 18.15, *Callister & Rethwisch 9e*. (Courtesy LaQue Center for Corrosion Technology, Inc.)

CORROSION PREVENTION (i)

- Materials Selection
  -- Use metals that are relatively unreactive in the corrosion environment -- e.g., Ni in basic solutions

  -- Use metals that passivate
  - These metals form a thin, adhering oxide layer that slows corrosion.



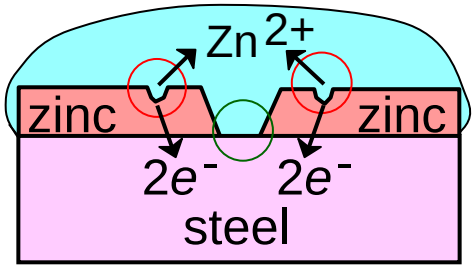
- Lower the temperature (reduces rates of oxidation and reduction)
- Apply physical barriers -- e.g., films and coatings

CORROSION PREVENTION (ii)

- Add inhibitors (substances added to solution that decrease its reactivity)
  -- Slow oxidation/reduction reactions by removing reactants (e.g., remove O_2 gas by reacting it w/an inhibitor).
  -- Slow oxidation reaction by attaching species to the surface.

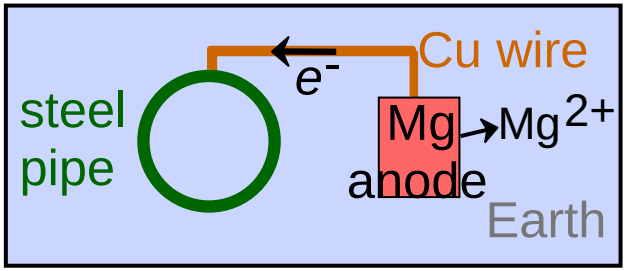
- Cathodic (or sacrificial) protection
  -- Attach a more anodic material to the one to be protected.

Galvanized Steel



e.g., zinc-coated nail

Using a sacrificial anode



e.g., Mg Anode

Adapted from Fig. 18.24, *Callister & Rethwisch 9e*.

Adapted from Fig. 18.23(a), *Callister & Rethwisch 9e*.

SUMMARY

- Metallic corrosion involves electrochemical reactions
  -- electrons are given up by metals in an oxidation reaction
  -- these electrons are consumed in a reduction reaction
- Metals and alloys are ranked according to their corrosiveness in standard emf and galvanic series.
- Temperature and solution composition affect corrosion rates.
- Forms of corrosion are classified according to mechanism
- Corrosion may be prevented or controlled by:
  -- materials selection
  -- reducing the temperature
  -- applying physical barriers
  -- adding inhibitors
  -- cathodic protection

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: